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International Finance

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Krone Currency Report

Analysis of the Economy

This currency report focuses on the Norwegian Krone (NOK), one of the most actively traded currencies in the world, managed by Norges Bank since 1816. Renowned for its stability, the Krone is often viewed as a safe-haven asset. Norway's strategic position along the North, Norwegian, and Barents Seas grants it access to significant oil reserves, positioning the country as a leading oil exporter. This resource wealth underpins the Government Pension Fund Global (GPFG)—the world's largest sovereign wealth fund—giving Norway extraordinary fiscal flexibility to stabilize its economy across business cycles and during economic crises.

In 2024, Norway's economy demonstrated modest growth, with overall GDP expanding by 2.1%, rebounding strongly from the 0.1% growth observed in 2023(Graph 1). This growth was primarily driven by robust oil and gas extraction activity. However, Mainland Norway's GDP, which excludes petroleum sectors, grew by only 0.6%, reflecting weak domestic demand. Investment levels (I) remained stagnant, with investment growth declining by 1.9% over 2024, signaling cautious business sentiment amid global uncertainties(Graph 4). Government spending (G) remained a pillar of support, accounting for 49.3% of GDP in 2024, up from 46.7% the previous year(Graph 5).

Household consumption (C) showed volatility following a sharp 24.3% decline in 2023; it dropped by 2.6% in 2024(Graph 6). Despite small recoveries in monthly figures toward the end of 2024, consumption levels remain well below pre-pandemic trends, reflecting the impact of high interest rates and inflation on household spending. Net exports (NX) contributed positively to GDP, with Norway recording a 2024 trade surplus of approximately 66.74 billion NOK. However, this represented a decrease from previous years due to falling energy prices (SSB).

Norway's economic strength was demonstrated during the COVID-19 pandemic. While many nations slashed rates to near-zero, and their economies languished, Norway initially dropped rates to 0% but quickly rebounded, aided by its sovereign wealth fund and a strong base of oil and machinery exports. This swift recovery allowed Norges Bank to begin raising rates earlier than other similar peers, resulting in a gradual rise in the policy rate to 4.5% by December 2024. However, Norges Bank opted to maintain the 4.5% rate in March 2025, despite inflation ticking back up to 3.6% in February from 2.3% in January (Graph 7), due to the possibility of signs of re-acceleration of inflation due to the violitity of the current global financial world and/or continual annual Increases in wage.

The recent drop in global oil prices—a critical export for Norway—has raised concerns about future growth prospects. As oil revenue weakens, pressure may mount on Norges Bank to shift toward monetary easing, potentially cutting rates later in 2025 to support domestic demand. Norway operates under a floating exchange rate regime, which allows the Krone to adjust to market forces without fixed intervention, while Norges Bank targets a 2% inflation rate. The sovereign wealth fund provides an additional buffer, allowing flexible fiscal responses and stabilization when needed.

Recent Trends of the Currency

The Norwegian Krone experienced notable depreciation in 2024, falling by 10.56% against the U.S. dollar. The Real Effective Exchange Rate (REER) stood at 73.9 (2005=100) in 2024(Graph 3), suggesting a weaker-than-average Krone historically. There are several factors contributing to this depreciation, including Norges Bank maintaining a high policy rate, there's also falling global oil prices reduced Norway’s export revenues, weighing further on the Krone, and global economic slowdowns among Norway’s major trading exports decreased prices possibly due to a global shift towards more eco-friendly energy sources when viewing it on a more broad scale, and looking at the rise in green energy use in the EU, who are large importers of norways oil.

Purchasing Power Parity (PPP) Analysis

Using the Big Mac Index, a Big Mac cost $7.14 in Norway in 2024, compared to $5.69 in the U.S. Using the Big Mac Index, the Krone appears to be overvalued by 25.47% compared to the U.S. dollar in 2024. However, relative PPP analysis based on CPI data shows that Norway’s inflation rate (2.6%) rose less than the U.S. (3.2%) over the same period (Graph 8), suggesting the Krone could modestly strengthen relative to the dollar in the medium term. When looking at the Big Mac Index, you can see the majority of Norwegian countries at the top of the list, pointing to the fact that other factors may be at play besides just the overvaluation of the Krone. Things like strong labor laws and structural price differences could also contribute to Norway's elevated Big Mac Index value. Again, pointing to the fact that the Big Mac Index is not a great signifier of PPP.

Macroeconomic Policy Analysis

Norges Bank’s restrictive stance in 2024 was aimed at controlling inflation (2% target) while maintaining economic growth. However, with oil prices falling and external demand weakening, the Bank has hinted at possible rate cuts later in 2025 to sustain growth momentum. Due to the initial economic boom from the pandemic, resulting in high levels of inflation and in turn, high interest rates. Government spending was projected at NOK 409.8 billion in 2024, representing 10.3% of Mainland Norway’s trend GDP. The sovereign wealth fund provides flexibility to fund expansionary measures without destabilizing public finances. Norway's monetary tightening and fiscal expansion have pulled the economy in opposite directions. Monetary policy restrained growth through higher interest rates. Tight monetary policy shifted the LM curve leftward, raising interest rates and suppressing output. Meanwhile, money demand shifted rightward as households held more cash amid rising uncertainty in Norway and the global Economy, and looking at the FX model, domestic returns also shifted up(Graph 9). showing the appreciation of the Krone against the US Dollar.

Time-Series Analysis

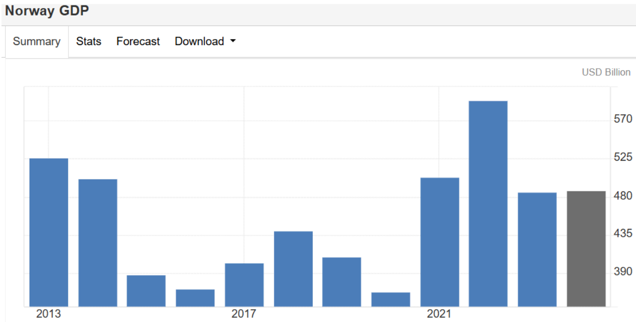
An ARIMA(0,1,1) model was developed using the Real effective exchange rate, Norway / Broad basket from 1995 to 2025, from the BIS. Initially, a Dickey-Fuller test revealed that the REER series was non-stationary (Graph 15). First differencing rendered the data stationary, as shown in the second Dickey-Fuller test (Graph 16). Partial autocorrelation analysis (PACF; Graph 13) indicated a strong lag at 1, suggesting an AR(1) component. The autocorrelation analysis (ACF; Graph 14) showed the moving average (MA) component to be more dominant at lag 1. Because of that, an ARIMA(0,1,1) model was selected. The ARIMA(0,1,1) model is statistically cleaner, eliminating the unnecessary AR term. Although the log-likelihood was slightly lower than ARIMA(1,1,1), the simpler model had stronger and more statistically significant parameters, so I opted to use that model. Forecasts using the ARIMA(0,1,1) model forecast continued depreciation pressure in the short, medium, and long term(Graphs 12, 11, & 10). I believe the short-term model's depreciation is due to weaker oil revenues and broader global economic uncertainty. The long-term forecast depreciation is likely due to a global trend in oil, and its perception, as the global economy gradually continues to shift away from using additional levels of brown energy(the Oil price has still not returned to 2013/2014 levels). If more aspects were taken into account, we would see stabilization or perhaps appreciation, as the inevitable decline is known to Norway, and steps to account for this would happen by diversifying their wealth fund more or something similar. The medium-term prediction, I believe, is a mixture of the two reasons listed above. While stabilization would be greatly reliant on a successful domestic policy pivot and recovery of external demand. With the supposed drop in interest rates, which would further drop the REER value, Norway would need to counteract that by cutting government spending, increasing taxes, or, more likely, using its sovereign wealth fund to buy more Krone.

Conclusion and Recommendation:

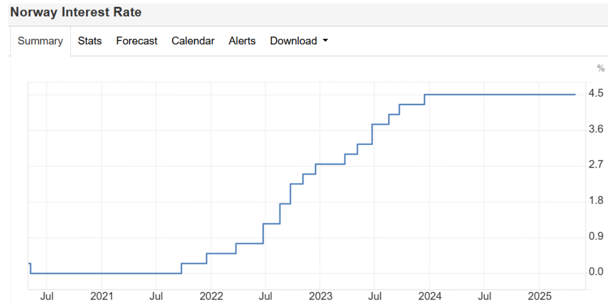
Norway's economy has historically shown strong resilience, underpinned by vast oil resources and the world's largest sovereign wealth fund. However, declining oil prices, slowing global demand, and a potentially necessary shift toward monetary easing introduce new risks for the Krone.

In the short run, a cautious approach toward NOK assets is advised. However, the Krone remains fundamentally strong in the medium to long term and represents a potentially attractive opportunity, based on its history and other assets, once oil markets and domestic consumption stabilize.

Appendix

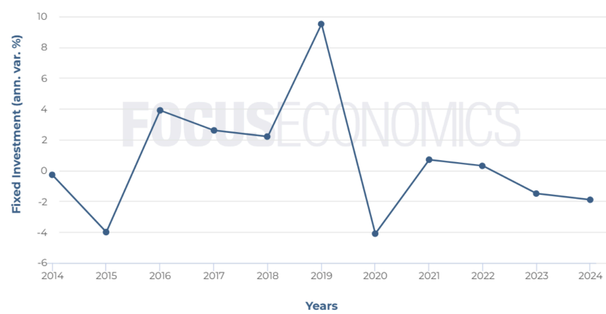


(Graph 1 - GDP)

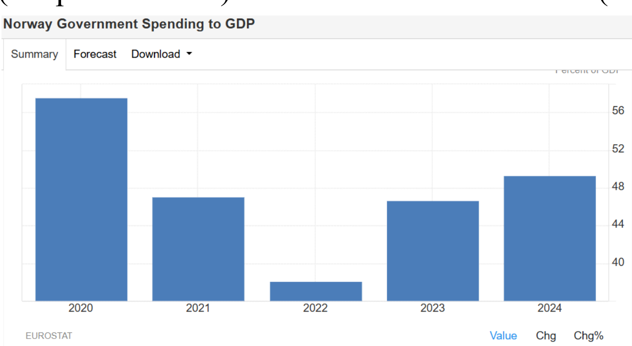


(Graph 2 - Interest Rate)

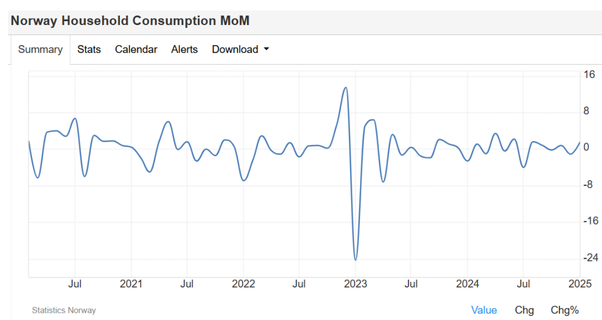
(Graph 3 - REER)



(Graph 4 - Investment)

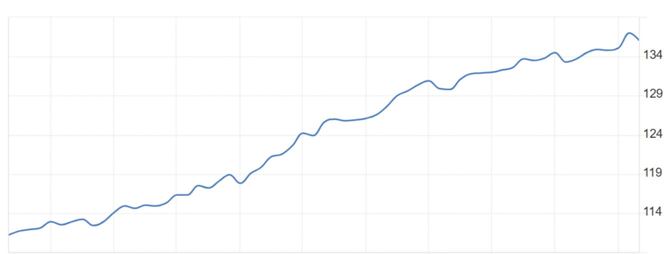


(Graph 5 - Gov spending)

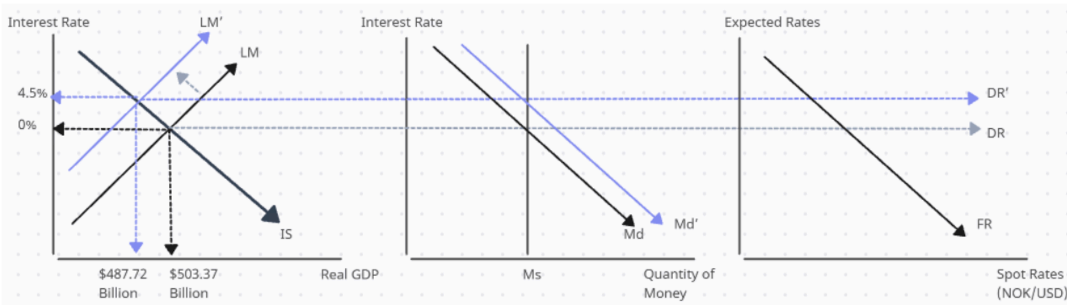


(Graph 6 - Consumption)

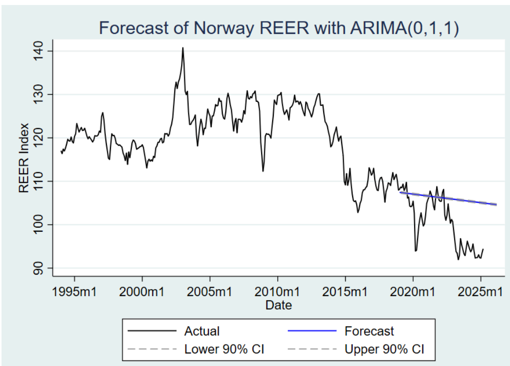
(Graph 7 - Inflation rate)

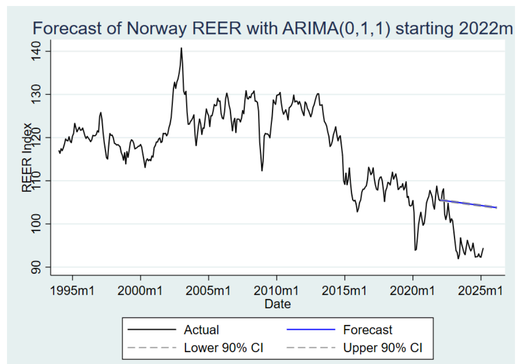


(Graph 8 - CPI)

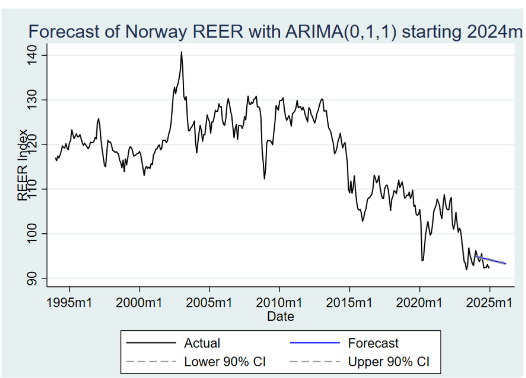


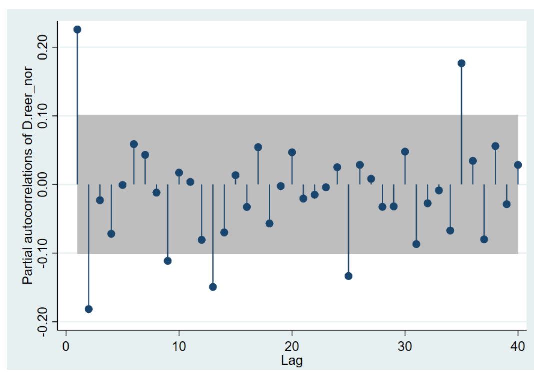
(Graph 9 IS/LM & Ms/Md & FRDR model)



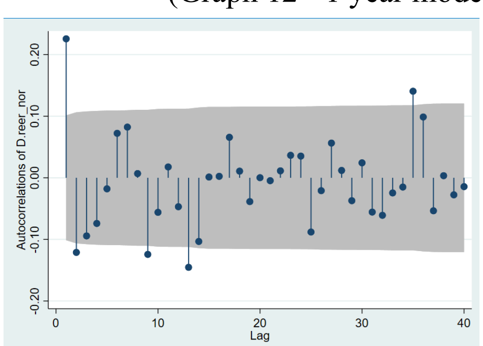


(Graph 10 - 5 year model) (Graph 11- 3 year model)





(Graph 12 - 1 year model) (Graph 13 - Partial Autocorrelation Function)

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. dfuller reer_nor

Dickey-Fuller test for unit root          Number of obs  = 374
Variable: reer_nor                        Number of lags =   0

H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                   Test          ────── critical value ──────
              statistic           1%           5%          10%

 Z(t)            -0.913       -3.450       -2.875       -2.570

MacKinnon approximate p-value for Z(t) = 0.7836.
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```
. dfuller d.reer_nor

Dickey-Fuller test for unit root          Number of obs  = 373
Variable: D.reer_nor                      Number of lags =   0

H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                   Test          ────── critical value ──────
              statistic           1%           5%          10%

 Z(t)           -15.302       -3.450       -2.875       -2.570

MacKinnon approximate p-value for Z(t) = 0.0000.
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(Graph 14 - Autocorrelation Function) (Graph 15 - Dickey-Fuller) (Graph 16 - Dickey-Fuller test on 1st difference)

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